

Modular Arithmetic

Did you know that you use modular arithmetic daily?

It's true!

The Clock

Every time you think about “time,” you use modular arithmetic because it deals with cycles of integers and remainders just like a clock.

For example, suppose your clock reads 9:00 (am/pm is not important).

What will the clock show in 10 hours?

Well, $9 + 10 = 19$, but “19 o'clock” is not something that can be displayed on a clock with the numbers 1 to 12.

So, what do we do?

We subtract 12 from 19 and proudly say that the clock will show 7:00.

This is the idea behind modular arithmetic, which is sometimes referred to as “clock arithmetic” because $19 \bmod 12 = 7 \bmod 12$, where 7 represents the remainder when 19 is divided by 12. You can review more history behind the idea at the [Institute for Advanced Studies](#).

How To Do Modular Arithmetic

This means that modular arithmetic finds the remainder of a number upon division!

Example #1

What is $16 \bmod 12$?

Well 16 divided by 12 equals 1 remainder 4. So the answer is 4!

Example #2

What about $15 \bmod 2$?

Here, 15 divided by 2 equals 7 remainder 1, so the solution is 1!

Example #3

And if you have $18 \bmod 9$?

Alright, so we know that 18 divided by 9 equals 2 remainder 0, so that means $18 \bmod 9$ is equivalent to 0!

So, what have we learned?

The trick for modular arithmetic is to focus on the remainder!

But just like we say with [divisibility](#), the remainder must be positive.

Example #4

For this problem, suppose we wanted to evaluate $-97 \bmod 11$.

Well, -97 divided by 11 equals -8 remainder -9.

But since this remainder is negative, we have to increase our quotient by 1 to say -97 divided by 11 equals -9 remainder 2, as $11(-9) + 2 = -97$!

Therefore, $-97 \bmod 11$ equals 2!

Modular Congruence

Now, in [number theory](#), we often want to focus on whether two integers say a and b , have the same remainder when divided by m . This is the idea behind modular congruence.

Congruence is nothing more than a **statement about divisibility** and was first introduced by Carl Friederich Gauss.

He surmised that if a and b are integers, and m is a positive integer, then a is congruent to b modulo m if and only if m divides a minus b . In other words, a congruence modulo asks whether or not a and b are in the same equivalence class.

If $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$, then $a \equiv b \pmod{m}$ if and only if $m \mid (a - b)$

Congruence Modulo Property

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Equivalence Class

What's an equivalence class?

It's best explained with an example.

Suppose we want to find the equivalence classes of mod 3. Well, we know that “**mod**” means we are interested in **remainders**, so what are the possible remainders when a number is divided by 3?

It's 0, 1, or 2!

Here's how.

Okay, so the remainders of 0, 1, or 2 comprise the equivalence classes for mod 3, which we write as [0], [1], and [2].

But let's look a bit deeper and find all the values within each class (partition).

Here's a trick, start by listing the equivalence classes as separate columns and then start at zero and keep writing numbers consecutively, wrapping to the next row, as shown below.

[0]	[1]	[2]
0	1	2
3	4	5
6	7	8
9	10	11
⋮	⋮	⋮

Equivalence Class Mod 3

This means that $[0] = \{\dots-6,-3,0,3,6,9,\dots\}$ and $[1] = \{\dots-5,-2,1,4,7,10,\dots\}$ and $[2] = \{\dots-4,-1,2,5,8,11,\dots\}$

Why?

Because every integer within the equivalence class has the same remainder when divided by 3!

Notice the following:

- $6/3 = 2$ with remainder 0.
- $7/3 = 2$ remainder 1.
- $8/3 = 2$ remainder 2.

Congruence Modulo — Examples

So, now let's see how equivalence classes help us determine congruence.

If n is a positive integer then integers a and b are congruent modulo n if they have the same remainder when divided by n . Another way to think of congruence modulo, is to say that integers a and b congruent modulo n if their difference is a multiple of n .

For example, 7 and 4 are congruent modulo 3 because not only are they in the same equivalence class, but their difference $7-4 = 3$, is a multiple of 3 (i.e., 3 divides 3), as shown below.

Problem #1

True or False: $7 = 4 \pmod{3}$

Comparing Equivalence Classes	Congruence Modulo Definition
$[1]$ 1 4 7 10 $\vdots$	$7 = 4 \pmod{3}$ $3 \mid (7 - 4)$ $3 \mid 3$

7 and 4 are in the same equivalence class, so they have the same remainder when divided by 3

3 divides 3 evenly

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Determine Congruence — Example

But 11 and 6 are not congruent modulo 3 because they are not in the same equivalence class and their difference, $11 - 6 = 5$, is not a multiple of 3, as illustrated below.

Problem #2

True or False: $11 \equiv 6 \pmod{3}$

	Comparing Equivalence Classes		Congruence Modulo Definition
	$[0]$	$[2]$	$11 \equiv 6 \pmod{3}$
	0	2	$3 \mid (11 - 6)$
	3	5	$3 \mid 5$
11 and 6 are in NOT in the same equivalence class, so they do NOT have the same remainder when divided by 3	6	8	3 does NOT divide 5 evenly
	9	11	
	$\vdots$	$\vdots$	

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Verify Incongruence — Example

So, what have we learned? If the remainders are equal to each other, then they are congruent! And the following theorem highlights this very idea:

Let $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$, then $a \equiv b \pmod{m}$ if and only if $a \bmod m = b \bmod m$

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Congruence Property

And this leads us to several vital theorems for congruence modulo.

Let $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$

If $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$

Symmetric Property

If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$

Transitive Property

If $a \equiv b \pmod{m}$ then $ac \equiv bc \pmod{mc}$ for $c > 0$

Scalar Multiple

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then $a + c \equiv (b + d) \pmod{m}$

Addition Property

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then $ac \equiv bd \pmod{m}$

Multiplication Property

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then $a - c \equiv (b - d) \pmod{m}$

Subtraction Property

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Modular Arithmetic Properties

In our lesson, we will work through several proofs for these essential theorems, as they are pivotal in our understanding of equivalence relations (future lesson) and allow us to define arithmetic operations.

For example, let's find:

1. $(7+9) \pmod{11}$
2. $(7)(9) \pmod{11}$

$7+9(\text{mod } 11)$	$7 \cdot 9(\text{mod } 11)$
$16(\text{mod } 11) \rightarrow 16 \div 11 = 1 \text{ remainder } 5$	$63(\text{mod } 11) \rightarrow 63 \div 11 = 5 \text{ remainder } 8$
$7+9(\text{mod } 11) \equiv 5(\text{mod } 11)$	$7 \cdot 9(\text{mod } 11) \equiv 8(\text{mod } 11)$
$7+9(\text{mod } 11) = 5$	$7 \cdot 9(\text{mod } 11) = 8$

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Modular Arithmetic Multiplication Addition

Arithmetic Modulo

And this leads us to Arithmetic Modulo m , where we can define arithmetic operations on the set of non-negative integers less than m , that is, the set $\{0, 1, 2, \dots, m-1\}$.

Addition on $\mathbb{Z}_m$, denoted $+_m$: $a +_m b = (a + b) \bmod m$

Multiplication on $\mathbb{Z}_m$, denoted $\cdot_m$: $a \cdot_m b = (a \cdot b) \bmod m$

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Arithmetic Modulo M Properties

The definition of **addition and multiplication modulo** follows the same properties of ordinary addition and multiplication of algebra.

The following properties are all satisfied.

- Closure
- Commutativity
- Associativity
- Identity
- Inverse
- Distributivity